SUBJECTIVE WORKSHEET

Heat

Science — Physics · Class 7 · Max Marks: 34 · Time: 1 hour

Section A — Very Short Answer (1 × 5 = 5)

1. Name the instrument used to measure temperature. [1]

2. State the normal temperature of the human body in °C. [1]

3. Name the mode of heat transfer that does not need a medium. [1]

4. Give one example of a good conductor of heat. [1]

5. Why should a clinical thermometer never be used to measure the temperature of boiling water? [1]

Section B — Short Answer (2 × 5 = 10)

1. Give any two differences between a clinical thermometer and a laboratory thermometer. [2]

2. What is conduction? Give one everyday example. [2]

3. Explain how a sea breeze is set up during the day. [3]

4. Why are the handles of cooking utensils usually made of wood or plastic? [2]

5. Why does a person wearing dark clothes feel hotter in the Sun than one wearing white clothes? [2]

Section C — Long Answer (5 × 3 = 15)

1. Explain the three modes of heat transfer, giving one example of each. [5]

2. Describe a simple activity to show that different solids conduct heat at different rates. [5]

3. **Explain, with the help of a diagram, the formation of a land breeze at night. [5]**

Section D — Higher-Order Thinking (Give Reasons) (2 × 2 = 4)

1. A block of ice is wrapped in a woollen blanket. Explain whether it melts faster or slower, and why. [2]

2. Two identical vessels, one black and one white, are filled with equally hot water. Which cools faster, and why? [2]

Answer Key

Very Short Answer

1. A thermometer.
2. About 37°C (98.6°F).
3. Radiation.
4. Any metal, e.g. copper or aluminium.
5. Its range only goes up to about 42°C, so the boiling water would overheat it and the thermometer would break.

Short Answer

1. Clinical: range 35–42°C, has a kink, used for body temperature. Laboratory: range about -10°C to 110°C, no kink, used for general lab measurements.
2. Conduction is the transfer of heat through a material from particle to particle without the particles moving along, e.g. a metal spoon getting hot in hot tea.
3. By day the land heats up faster than the sea; the warm air over the land rises and the cooler air from over the sea moves in to take its place — this is the sea breeze.
4. Wood and plastic are poor conductors (insulators) of heat, so the handle stays cool and does not burn the hand.
5. Dark surfaces absorb most of the heat radiation, while white surfaces reflect most of it, so dark clothes feel hotter.

Long Answer

1. Conduction — heat passes through solids from particle to particle (metal spoon in hot tea). Convection — the heated liquid/gas itself moves (boiling water, sea breeze). Radiation — heat travels without any medium (the Sun's heat, warmth near a fire).
2. Fix small wax balls with equal wax at the ends of rods of copper, aluminium, iron and glass. Heat the other ends together. The wax ball on the best conductor (copper) falls first and the one on glass falls last, showing metals conduct heat faster than glass.
3. At night the land cools faster than the sea. The warmer air over the sea rises and the cooler air from over the land moves out towards the sea to replace it — this flow of air from land to sea is the land breeze. (Diagram: arrows from land to sea, warm air rising over the sea.)

Higher-Order Thinking (Give Reasons)

1. It melts slower. Wool (and the air trapped in it) is a poor conductor, so it slows the flow of heat from the warm surroundings into the ice.
2. The black vessel cools faster, because dark surfaces are better emitters (radiators) of heat than white surfaces.